

# Where Our Plastic Comes From, and the Case for Cost-Recovery

Saint Lucia — Eastern Caribbean (Lesser Antilles)

Prepared 2026-06-29 · oceanplastic v0.1 (MVP)

**DRAFT · FIGURES ARE ILLUSTRATIVE**

## ⚠ WHAT THIS REPORT IS — AND IS NOT

As of mid-2026 there is no operative international mechanism that compels a polluting state or company to pay a coastal country for beached plastic. This document is therefore an evidence-and-leverage instrument — for domestic Extended Producer Responsibility (EPR) cost-recovery, treaty/regional-seas advocacy, and reputational pressure — not a binding invoice. Its credibility is its candor about what it cannot prove.

HOW TO READ EVERY CLAIM

**A** **Defensible** — producer of a branded item; material; item sector. **B** **Probabilistic** — source region / emission pathway (disclose uncertainty). **C** **Never asserted** — the emitting country of a specific item; a single origin.

1 Observed evidence on our shore

74 geotagged litter observations are documented along this coastline across 2 clusters (source: **openlittermap**, licence ODbL-1.0). This is a *documented sample*, not a census of everything that has washed up.

2 Producer responsibility

TIER A · CORPORATE

Computed from the Break Free From Plastic brand-audit dataset (Cowger et al. 2024, CC-BY-4.0) at the **regional** level. Shares are of branded (attributable) items.



MANDATORY DISCLOSURES

- ~63% of audited items are UNBRANDED — the producer axis is silent on them. Shares above are of the branded remainder only.
- Producer attribution is COUNT-BASED, not mass-based; rankings shift with audit location and ranking rule (volume vs geographic spread).
- Brand strings are collaborator-entered and not fully standardized (BFFP data note).
- Producer shares were computed at the 'regional' level. Producer shares computed from 63 brand-audit events in 4 Caribbean-basin countries (Saint Lucia has no BFFP audits).

This axis names companies, never countries. It does not say any nation polluted Saint Lucia.

3 Emission pathway & source region

TIER B · PROBABILISTIC

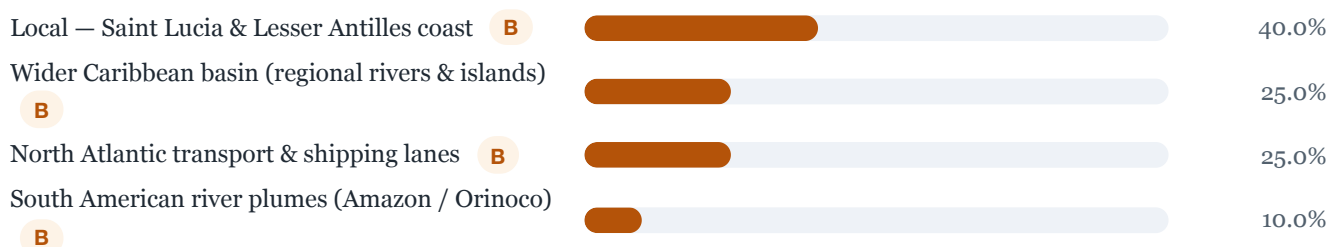
ILLUSTRATIVE PRIOR — NOT A CALIBRATED MODEL

Tier B — ILLUSTRATIVE PRIOR, not a calibrated drift ensemble. Pathway split is the Meijer & van Sebille (2021) backtracking posterior; the region split is an analyst prior from the pilot config. Replace with an OpenDrift backtracking ensemble before relying on it.

## Likely emission pathway



## Illustrative source-region prior for this basin



## 4 What kind of plastic

TIER A · COMPOSITION ONLY

| By sector / use     | Share | Tier | By material / polymer | Share | Tier |
|---------------------|-------|------|-----------------------|-------|------|
| food packaging      | 33.5% | A    | pet                   | 29.6% | A    |
| other               | 7.3%  | A    | o                     | 22.2% | A    |
| household products  | 3.3%  | A    | ps                    | 15.9% | A    |
| smoking materials   | 2.6%  | A    | pp                    | 12.0% | A    |
| packaging materials | 2.1%  | A    | hdpe                  | 3.2%  | A    |
| personal care       | 1.0%  | A    | ldpe                  | 1.5%  | A    |

Polymer is material type only — it carries zero geographic provenance.

### THE GOVERNING CONFOUND — MANUFACTURED-IN ≠ CONSUMED-IN ≠ EMITTED-FROM

Where audited litter was *collected* (consumed-in) is not where it was made or emitted:

- Mexico — 72.4% of audited items
- Colombia — 17.9% of audited items
- Jamaica — 9.6% of audited items
- Puerto Rico — 0.0% of audited items

## 5 The monetary ask — banded, never point estimates

Estimated documented-sample mass: **0.4 – 2.2 kg (central 0.9 kg)**. Illustrative coastline-scaled mass: 1.2 – 111.0 kg (central 8.9 kg).

| Layer                                           | Estimate (USD)                                    | Role        | Basis                                                                                  |
|-------------------------------------------------|---------------------------------------------------|-------------|----------------------------------------------------------------------------------------|
| Local cleanup cost (documented sample)          | \$0 – \$6 (central \$1)                           | HEADLINE    | 74 documented OpenLitterMap observations x 5.0-30.0 g/item x \$0.5-2.5/kg cleanup.     |
| Cleanup cost scaled to coastline (ILLUSTRATIVE) | \$1 – \$278 (central \$11)                        | CONTEXT     | Documented sample / sample_fraction (0.02-0.3).                                        |
| Ecosystem-service / natural-capital value       | \$1 – \$73 (central \$9)                          | CONTEXT     | Documented mass x UNEP/GESAMP \$3,300-33,000 per tonne reduced marine natural capital. |
| Full-lifecycle social cost                      | \$7 – \$333 (central \$53)                        | COMPARATOR  | Documented mass x WWF/Dalberg \$19-150 per kg lifecycle social cost.                   |
| Plastic-credit price (cross-check)              | \$0 – \$2 (central \$0)                           | CROSS-CHECK | Documented mass x World Bank/Verra \$140-800 per tonne plastic-credit range.           |
| Tourism / fisheries revenue at risk             | \$5,500,000 – \$55,000,000 (central \$22,000,000) | HEADLINE    | \$1,100,000,000 annual tourism receipts x 0.5%-5.0% debris deterrence.                 |

#### HEADLINE ASK – DEFENSIBLE FLOOR (LAYER 1 + LAYER 2)

**\$5,500,000 – \$55,000,006 (central \$22,000,001)**

Worked corporate example – **the coca-cola company** at 12.2% of all documented items → attributable share ≈ **\$670,786 – \$6,707,857 (central \$2,683,143)** (an EPR/contribution figure, not a court award).

Per-layer caveats

- **Local cleanup cost (documented sample)** (headline): Bottom-up local cost; replace the illustrative per-item mass and \$/kg with measured local figures. Covers the documented sample only.
- **Cleanup cost scaled to coastline (ILLUSTRATIVE)** (context): Depends entirely on the assumed sampling fraction — illustrative, NOT a headline figure. A defensible report measures coastline density directly.
- **Ecosystem-service / natural-capital value** (context): Willingness-to-avoid valuation (2011-era), not an auditable invoice. Context anchor.
- **Full-lifecycle social cost** (comparator): Bundles GHG + health + waste externalities across the WHOLE lifecycle; overstates what is attributable to this coastline. Comparator only.
- **Plastic-credit price (cross-check)** (cross-check): Credits are contested (GAIA/BFFP greenwashing findings: additionality failures, incineration). Never the core ask; require registry-listed + audited + retired provenance.
- **Tourism / fisheries revenue at risk** (headline): Case-backed (e.g. Cape Town, Zanzibar); deterrence % is illustrative — calibrate locally.

Uncertainty context: bottom-up river models put global input at 1.1 – 2.4 Mt/yr (central 1.8 Mt/yr) (Lebreton 2017), while the top-down seawater constraint is 0.1 – 3.8 Mt/yr (central 0.7 Mt/yr) (Strokal 2023) — cited so this report is not accused of cherry-picking the high end.

6 Where this goes & the lever it pulls

Primary forum: **domestic\_epr** · jurisdiction: **Saint Lucia** · regional: Cartagena Convention (Wider Caribbean Region).

| Forum                                                        | Principle                                         | Enforceability                                                                                                                                        | Role          |
|--------------------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Domestic EPR regulator</b>                                | Extended Producer Responsibility                  | HARD / enforceable — the only genuinely binding cost-recovery lever, but reaches only producers within your own jurisdiction.                         | Lead ask      |
| <b>UNEA / INC Global Plastics Treaty</b>                     | Polluter-pays + loss-and-damage framing           | Soft / advocacy — negotiations deadlocked (INC-5.1/5.2/5.3); draft finance is implementation funding, not liability/compensation.                     | Advocacy      |
| <b>Regional Seas (Cartagena Convention, Wider Caribbean)</b> | Cooperation + regional litter plans               | Mixed — concrete forum, some binding plans, but no victim-compensation machinery.                                                                     | Diplomacy     |
| <b>Basel Convention</b>                                      | Prior Informed Consent on waste exports           | Binding but wrong remedy — lets you refuse/control imports, gives no right to be paid.                                                                | Leverage      |
| <b>IMO MARPOL Annex V</b>                                    | Ban on ship-source plastic discharge              | Binding but flag/port-state enforced — no coastal-state compensation right.                                                                           | Leverage      |
| <b>UNCLOS Part XV / ITLOS</b>                                | No-transboundary-harm duty + state responsibility | Hard in theory, UNTESTED for plastics; consent-based jurisdiction. 2024 ITLOS / 2025 ICJ advisory opinions are tailwinds but advisory and about GHGs. | Appendix only |

**Framing:** lead with the enforceable EPR cost-recovery ask; present the regional-seas / UNEA asks as advocacy leverage; reserve the UNCLOS pathway as a credible-but-untested appendix.

## A Global emission baseline — context, not blame

Top mismanaged-plastic emitters (Jambeck 2015 / Our World in Data, 2010 base year). Global context — *not* a claim about who polluted Saint Lucia; the basin drift prior in §3 speaks to this shoreline.

| Country     | Mismanaged plastic (Mt/yr, 2010) | Share of listed |
|-------------|----------------------------------|-----------------|
| China       | 8.82                             | 35.4%           |
| Indonesia   | 3.22                             | 12.9%           |
| Philippines | 1.88                             | 7.5%            |
| Vietnam     | 1.83                             | 7.3%            |
| Sri Lanka   | 1.59                             | 6.4%            |
| Thailand    | 1.03                             | 4.1%            |
| Egypt       | 0.97                             | 3.9%            |
| Malaysia    | 0.94                             | 3.8%            |

## B Limitations & credibility risks

The things that could get this report dismissed — and how each is caveated so it survives scrutiny.

1. **Manufactured-in != emitted-from.** Producer responsibility and emission pathway are reported as SEPARATE axes; no emitting country is named from item-level evidence (the Inaccessible Island trap).
2. **Brand audits blind to ~half the items, count-not-mass.** The unbranded share is stated; counts and (where possible) mass kept separate; rankings noted to shift with audit location and ranking rule.
3. **River/country emission models weakly validated, disagree by ~2 orders of magnitude.** Figures are ranges, used as ordinal/relative; the Strokal-2023 top-down ~0.7 Mt/yr anchor is cited against the bottom-up centrals.
4. **Drift attribution is probabilistic, dominated by unknown floating time.** Region/pathway are an illustrative Tier-B prior (a STUB), never a country/river for a single item; the full build outputs a probability map with ensemble spread.
5. **Plastic-credit valuation invites greenwashing attack.** Credits are a cross-check only, never the core ask; GAIA/BFFP critiques disclosed.
6. **No enforceable forum exists.** Framed as EPR cost-recovery + a contribution case (the L&D no-liability template), not a verdict.
7. **Data-license / provenance traps.** License tracked per record; CC-BY-NC sources quarantined; commercial-safe sources preferred.
8. **Single-vendor / single-source figures.** Such figures are labelled; multiply-corroborated numbers (Jambeck headline, Cowger brand shares, World Bank credit range) are led with.

## C Data sources & licences

| Source                                                               | Role                 | Licence         | Commercial-safe |
|----------------------------------------------------------------------|----------------------|-----------------|-----------------|
| OpenLitterMap                                                        | item-observations    | ODbL-1.0        | yes             |
| Break Free From Plastic brand audit (Cowger et al. 2024)             | producer-attribution | CC-BY-4.0       | yes             |
| Lebreton et al. 2017 river plastic emissions                         | river-baseline       | CC-BY-4.0       | yes             |
| Jambeck et al. 2015 mismanaged plastic waste (via Our World in Data) | country-baseline     | CC-BY-4.0       | yes             |
| Strokal/Weiss et al. 2023 top-down seawater constraint               | uncertainty-anchor   | CC-BY-4.0       | yes             |
| Meijer et al. 2021 (via theoceancleanup.com framing)                 | river-baseline       | CC-BY-NC-ND-3.0 | quarantined     |
| Minderoo Global Plastic Watch (satellite)                            | land-leakage-context | CC-BY-NC-2.0    | quarantined     |
| Ocean Conservancy TIDES (International Coastal Cleanup)              | item-observations    | unresolved      | quarantined     |

Quarantined (CC-BY-NC / unresolved terms), used for context only: Meijer et al. 2021 (via theoceancleanup.com framing); Minderoo Global Plastic Watch (satellite); Ocean Conservancy TIDES (International Coastal Cleanup).

Generated by the oceanplastic MVP. Replace illustrative defaults (per-item mass, \$/kg, deterrence %, sampling fraction, drift prior) with locally-measured values before any real submission.